

Ethical Hacking – CO3 – AT1

Nmap Enumeration Commands

Q1. A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server `scanme.nmap.org`. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using Nmap, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on `scanme.nmap.org`.

1. Host discovery

```
nmap -sn scanme.nmap.org  
  
# Ping sweep / host discovery only, no port scan
```

2. TCP port scan (full range)

```
nmap -sS -p- scanme.nmap.org  
  
# SYN scan across all 65535 TCP ports
```

3. Service and version detection

```
nmap -sV scanme.nmap.org  
  
# Probe open ports to identify service/version
```

4. OS fingerprinting

```
nmap -O scanme.nmap.org  
  
# TCP/IP stack fingerprinting for OS guess
```

5. Default script scan

```
nmap -sC scanme.nmap.org  
  
# Run the default NSE script category
```

6. Combined enumeration (recommended single run)

```
nmap -sS -sV -O -sC -p- -T4 scanme.nmap.org -oN scanme_full_scan.txt  
  
# All checks above in one scan; -oN saves normal-format output for the report
```

Q2. The security team has deployed a deliberately vulnerable Linux server, 192.168.56.101 (Metasploitable 2), in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use Nmap to perform comprehensive enumeration of 192.168.56.101. Identify all open ports, running services, service versions, and operating system information.

1. Host discovery

```
nmap -sn 192.168.56.101  
  
# Confirm host is up in the isolated lab network
```

2. TCP port scan (full range)

```
nmap -sS -p- 192.168.56.101  
  
# SYN scan across all TCP ports
```

3. Service and version detection

```
nmap -sV 192.168.56.101  
  
# Identify running services and their versions
```

4. OS fingerprinting

```
nmap -O 192.168.56.101  
  
# Determine likely operating system
```

5. Default script scan

```
nmap -sC 192.168.56.101  
  
# Run default NSE scripts against open ports
```

6. Combined enumeration (recommended single run)

```
nmap -sS -sV -O -sC -p- -T4 192.168.56.101 -oN metasploitable2_scan.txt  
  
# Full enumeration in one pass, saved for the risk-categorization table
```

Q3. A financial organization wants to assess the security posture of a Windows workstation located at 192.168.56.102 before connecting it to the production network. Using Nmap, enumerate 192.168.56.102 to identify active ports, network services, file-sharing protocols, remote management services, and operating system details.

1. Host discovery

```
nmap -sn 192.168.56.102  
  
# Confirm host is reachable
```

2. TCP port scan (full range)

```
nmap -sS -p- 192.168.56.102  
  
# SYN scan across all TCP ports
```

3. Service and version detection

```
nmap -sV 192.168.56.102  
  
# Identify file-sharing / remote-management services
```

4. OS fingerprinting

```
nmap -O 192.168.56.102  
  
# Confirm Windows version/build family
```

5. Default script scan

```
nmap -sC 192.168.56.102  
  
# Run default NSE scripts against open ports
```

6. Targeted SMB / RDP enumeration

```
nmap --script smb-os-discovery,smb-enum-shares,smb-security-mode -p 139,445 192.168.56.102  
  
# Enumerate SMB shares, security mode, and OS details over file-sharing ports  
  
nmap --script rdp-enum-encryption -p 3389 192.168.56.102  
  
# Check RDP encryption/security settings if port 3389 is open
```

7. Combined enumeration (recommended single run)

```
nmap -sS -sV -O -sC -p- -T4 192.168.56.102 -oN windows_workstation_scan.txt  
  
# Full enumeration in one pass for the least-privilege / attack-surface analysis
```

Q4. An organization hosts a web application on 192.168.56.103 (OWASP Juice Shop Server) and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using Nmap, perform a detailed enumeration of 192.168.56.103, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification.

1. Host discovery

```
nmap -sn 192.168.56.103  
  
# Confirm host is reachable
```

2. TCP port scan (full range)

```
nmap -sS -p- 192.168.56.103  
  
# SYN scan across all TCP ports
```

3. Service and version detection

```
nmap -sV 192.168.56.103  
  
# Identify the web service/app server and version
```

4. OS fingerprinting

```
nmap -O 192.168.56.103  
  
# Determine host operating system
```

5. Default script scan

```
nmap -sC 192.168.56.103  
  
# Run default NSE scripts against open ports
```

6. Web service identification

```
nmap --script http-title,http-headers,http-enum -p 80,3000,8080 192.168.56.103  
  
# Identify page titles, server headers, and common web paths/admin interfaces
```

7. Combined enumeration (recommended single run)

```
nmap -sS -sV -O -sC -p- -T4 192.168.56.103 -oN juiceshop_scan.txt  
  
# Full enumeration in one pass to support the exposure/risk analysis
```

Q5. The network gateway device at 192.168.56.1 serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using Nmap, conduct a comprehensive enumeration of 192.168.56.1, including port scanning, service detection, operating system fingerprinting, and script-based enumeration.

1. Host discovery

```
nmap -sn 192.168.56.1  
  
# Confirm gateway is reachable
```

2. TCP port scan (full range)

```
nmap -sS -p- 192.168.56.1  
  
# SYN scan across all TCP ports
```

3. Service and version detection

```
nmap -sV 192.168.56.1  
  
# Identify management services (SSH, Telnet, HTTP/S, etc.)
```

4. OS fingerprinting

```
nmap -O 192.168.56.1  
  
# Determine gateway/router OS or firmware family
```

5. Default and management-focused script scan

```
nmap -sC 192.168.56.1  
  
# Run default NSE scripts against open ports  
  
nmap -sU -p 161 --script snmp-info,snmp-sysdescr 192.168.56.1  
  
# Check for exposed/misconfigured SNMP service (UDP)  
  
nmap --script ssh2-enum-algos -p 22 192.168.56.1  
  
# Enumerate supported SSH algorithms if port 22 is open  
  
nmap --script http-title,http-headers -p 80,443 192.168.56.1  
  
# Fingerprint the web-based management interface, if present
```

6. Combined enumeration (recommended single run)

```
nmap -sS -sV -O -sC -p- -T4 192.168.56.1 -oN gateway_scan.txt  
  
# Full enumeration in one pass to support the management-interface risk analysis
```

